

6. Lineární lomená funkce a mocninné funkce (MO 07)

nepřímá úměrnost jako funkce

lineární lomená funkce, definiční obor, obor hodnot, nalezení asymptot

průsečíky s osami souřadnic

mocninná funkce (kladný, záporný, sudý, lichý exponent, racionální exponent)

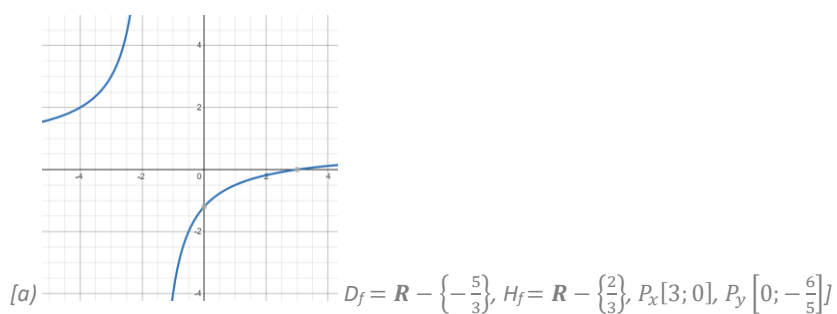
Teorie + tabulky:

Příklady, které mi nešly:

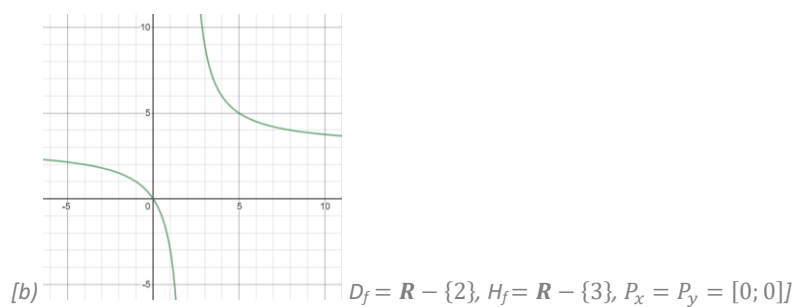
Co mi nejde + dotazy:

1. Sestrojte grafy funkcí, určete D_f , H_f a průsečíky s osami:

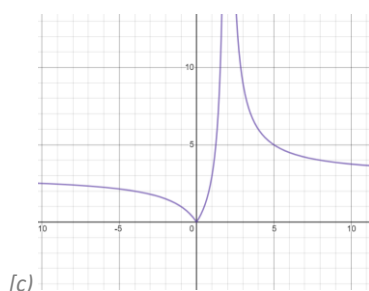
a) $y = \frac{2x-6}{3x+5}$



b) $y = \frac{3x}{x-2}$



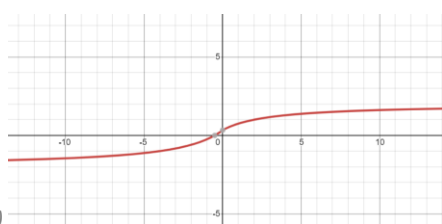
c) $y = \left| \frac{3x}{2-x} \right|$



[c]

$$D_f = \mathbf{R} - \{2\}, H_f = \langle 0; \infty \rangle, P_x = P_y = [0; 0]$$

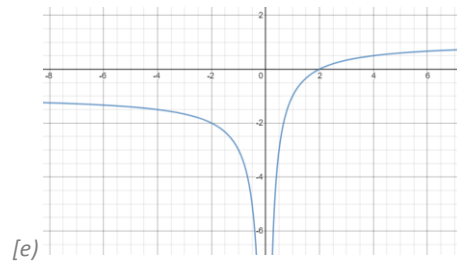
d) $y = \frac{2x+1}{|x|+3}$



[d]

$$D_f = \mathbf{R}, H_f = (-2; 2), P_x \left[-\frac{1}{2}; 0 \right], P_y \left[0; \frac{1}{3} \right]$$

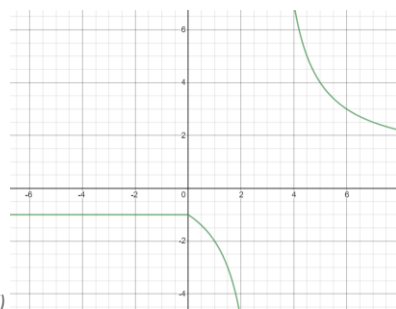
e) $y = \frac{x-2}{|x|}$



[e)

$D_f = \mathbf{R} - \{0\}$, $H_f = (-\infty; 1)$, $P_x[2; 0]$, P_y neex.]

f) $y = \frac{x+3}{|x|-3}$



[f)

$D_f = \mathbf{R} - \{\pm 3\}$, $H_f = (-\infty; -1) \cup (1; \infty)$, P_x neex., $P_y[0; -1]$

2. Načrtněte grafy funkcí:

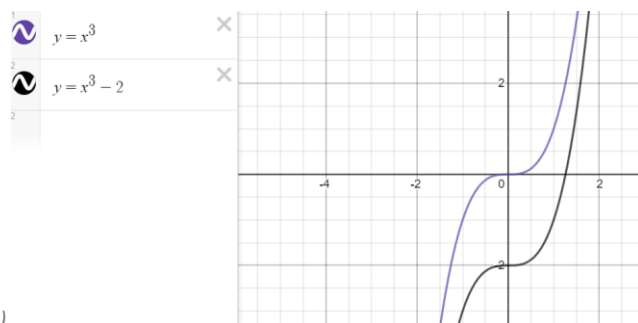
a) $y = x^3$

b) $y = x^3 - 2$

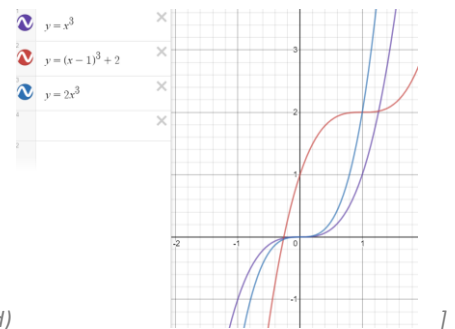
c) $y = (x - 1)^3 + 2$

d) $y = 2x^3$

[a) +b)



c)+d)



3. Vyberte sudé funkce a načrtněte jejich graf:

a) $y = x^{-3}$

b) $y = x^2 - 2$

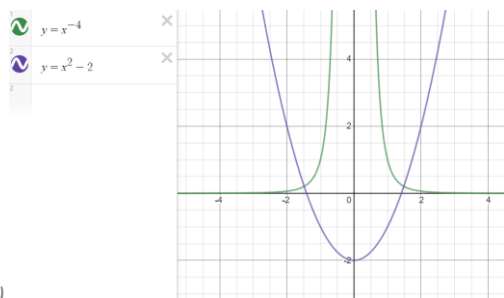
c) $y = x^4$

d) $y = 3x^4$

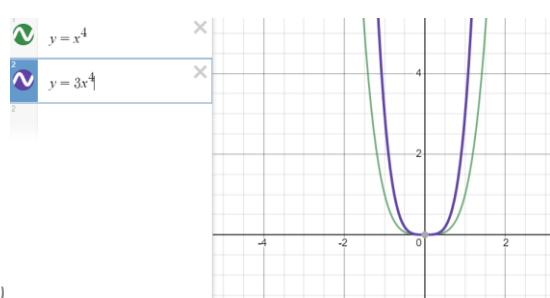
e) $y = x^5$

f) $y = (x - 1)^4$

g) $y = x^{-4}$



[sudé – b) + g)



; c) + d)

]

4. Vyberte liché funkce a načrtněte jejich graf:

a) $y = x^{-6}$

d) $y = 2x^5$

g) $y = x^{-5}$

b) $y = (x + 2)^{-3}$

e) $y = x^5$

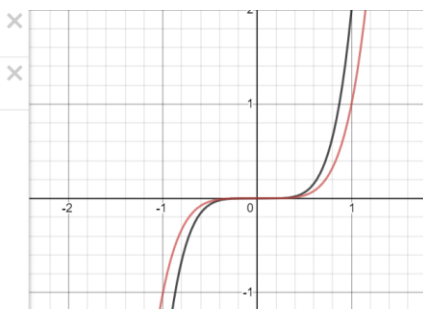
c) $y = x^4$

f) $y = x^{-3} - 1$

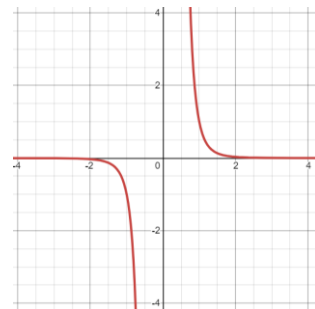
1 $y = 2x^5$

2 $y = x^5$

2



[liché – d) + e)



g)

Najděte k zadané funkci inverzní funkci. Načrtněte jejich graf a určete jejich definiční obor.

a) $y = \sqrt{x}$

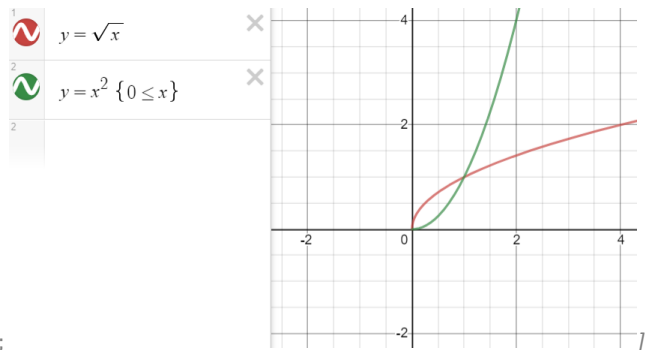
b) $y = \sqrt{x-1}$

c) $y = \sqrt{2x+3}$

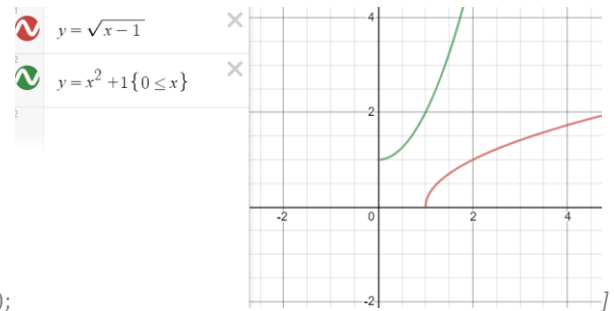
d) $y = \sqrt[3]{x}$

e) $y = \sqrt[3]{x+2}$

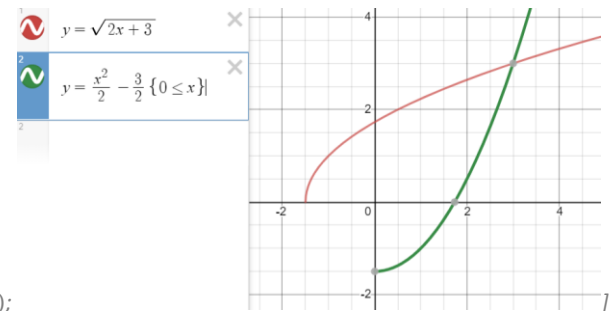
f) $y = \sqrt[3]{3x-4}$



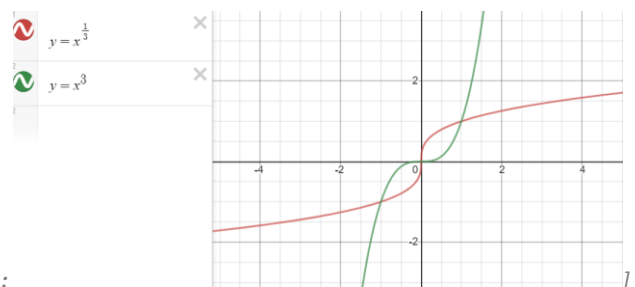
[a] $D(f) = \langle 0; \infty \rangle$; $f^{-1}: y = x^2$; $D(f^{-1}) = \langle 0; \infty \rangle$;



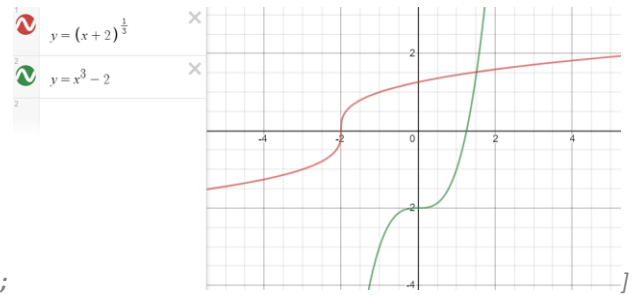
[b] $D(f) = \langle 1; \infty \rangle$; $f^{-1}: y = x^2 + 1$; $D(f^{-1}) = \langle 0; \infty \rangle$;



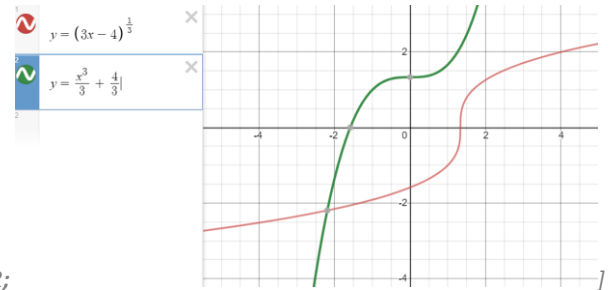
[c] $D(f) = \left\langle -\frac{3}{2}; \infty \right\rangle$; $f^{-1}: y = \frac{x^2}{2} - \frac{3}{2}$; $D(f^{-1}) = \langle 0; \infty \rangle$;



[d] $D(f) = \mathbf{R}$; $f^{-1}: y = x^3$; $D(f^{-1}) = \mathbf{R}$;



$$[e) D(f) = \mathbb{R}; f^{-1}: y = x^3 - 2; D(f^{-1}) = \mathbb{R};$$



$$[f) D(f) = \mathbb{R}; f^{-1}: y = \frac{x^3}{3} + \frac{4}{3}; D(f^{-1}) = \mathbb{R};$$

5. Určete definiční obor funkcí:

a) $y = \frac{3}{x^2-1}$

b) $y = \sqrt{x^2 - 3x + 2}$

c) $y = \frac{4-|x|}{3-|x|} - \sqrt{10-3x-x^2}$

$$[a) \mathbf{R} - \{-1, 1\} \quad b) (-\infty; 1) \cup (2; \infty) \quad c) \langle -5; 2 \rangle - \{-3\}]$$

6. Načrtněte grafy funkcí, určete jejich $D(f)$, $H(f)$, najděte souřadnice průsečíků grafu funkce se souřadnicovými osami, posuďte, zda jsou sudé nebo liché.

a) $y = |x|^{-3}$

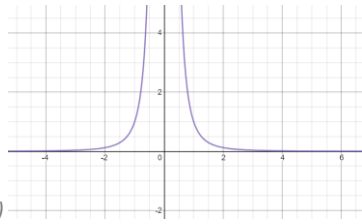
b) $y = (x - 1)^{-3}$

c) $y = |x^{-4}| + 1$

d) $y = -x^{-5}$

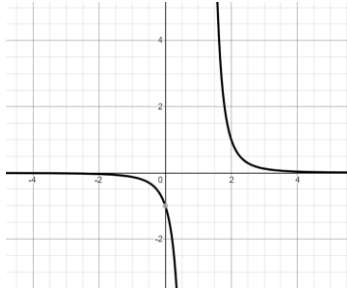
e) $y = |x^{-5}| - 1$

f) $y = |x^{-5} - 1|$



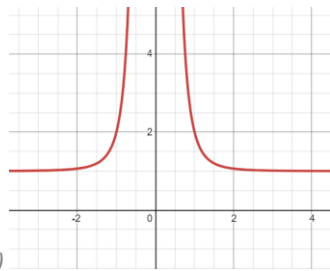
[a)

$$D(f) = \mathbf{R} - \{0\}; H(f) = (0; \infty); P_x, P_y \text{ neex.}; \text{sudá fce}]$$



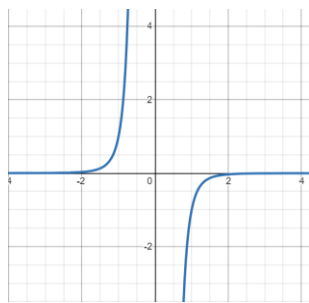
[b)

$$D(f) = \mathbf{R} - \{1\}; H(f) = \mathbf{R} - \{0\}; P_x \text{ neex.}, P_y [0; -1]; \text{ani sudá ani lichá fce}]$$



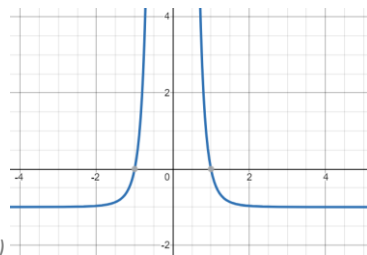
[c)

$$D(f) = \mathbf{R} - \{0\}; H(f) = (1; \infty); P_x, P_y \text{ neex.}; \text{sudá fce}]$$



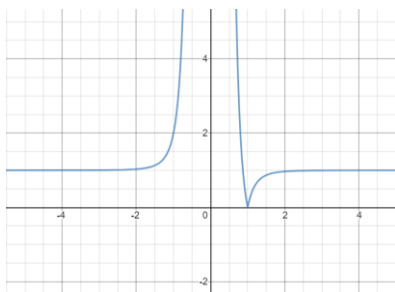
[d)

$$D(f) = \mathbf{R} - \{0\}; H(f) = \mathbf{R} - \{0\}; P_x, P_y \text{ neex.}; \text{lichá fce}]$$



[e)

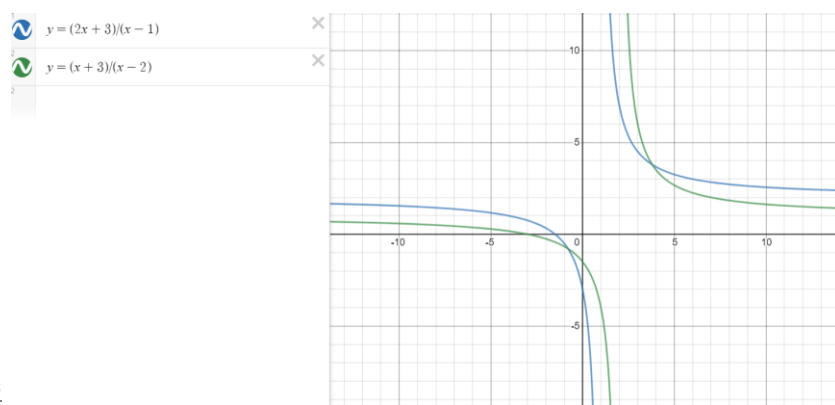
$$D(f) = \mathbf{R} - \{0\}; H(f) = (-1; \infty); P_x [\pm 1, 0], P_y \text{ neex.}; \text{sudá fce}]$$



[f)

$$D(f) = \mathbf{R} - \{0\}; H(f) = \langle 0; \infty \rangle; P_x [1, 0], P_y \text{ neex.}; \text{ani sudá ani lichá fce}]$$

7. Najděte předpis inverzní funkce k funkci $f: y = \frac{2x+3}{x-1}$ funkci. Obě funkce načrtněte. Napište rovnice jejich asymptot.



$$[f^{-1} : y = \frac{x+3}{x-2}]$$

as. $f: x = 1, y = 2$; as. $f^{-1} : x = 2, y = 1$

8. Vyjádřete objem krychle V (m^3) o hraně a (m) jako funkci její tělesové úhlopříčky u (m).

$$[V = \frac{u^3 \cdot \sqrt{3}}{9}]$$